Optical Networking
(Link Budget Design)

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Optical Power Budget

* BER is a key objective of the optical system design.
* Goal is to get from Tx to Rx with a BER < BER threshold of the Rx.
* BER thresholds are on data sheets.
* Typical minimum acceptable rate is 10^{-12}

The Optical Power Budget is :

Optical Power Budget = Power Sent –Receiver Sensitivity

Terminology—Fiber Impairments

- Attenuation = Loss of power in dB/km.
- Chromatic Dispersion = Spread of light pulse in ps/nm-km.
- Optical Signal-to-Noise Ratio (OSNR) = Ratio of optical signal power to noise power for the receiver.

Types of Single-Mode Fiber

- SMF (standard, 1310 nm optimized,
  - Most widely deployed so far, introduced in 1986, cheapest
- DSF (Dispersion Shifted, G.653)
  - Intended for single channel operation at 1550 nm
- NZDSF (Non-Zero Dispersion Shifted, G.655)
  - For WDM operation in the 1550 nm region only
  - TrueWave™, FreeLight™, LEAF, TeraLight™, etc.
    - Latest generation fibers developed in mid 90s
    - For better performance with high capacity DWDM systems
  - MetroCor™, WideLight™
    - Low PMD ultra long haul fibers

TrueWave Is a Trademark of Lucent; TeraLight Is a Trademark of Alcatel;

FreeLight and WideLight Are Trademarks of Pirelli; MetroCor Is a Trademark of Corning

Different Solutions for Different Fiber Types

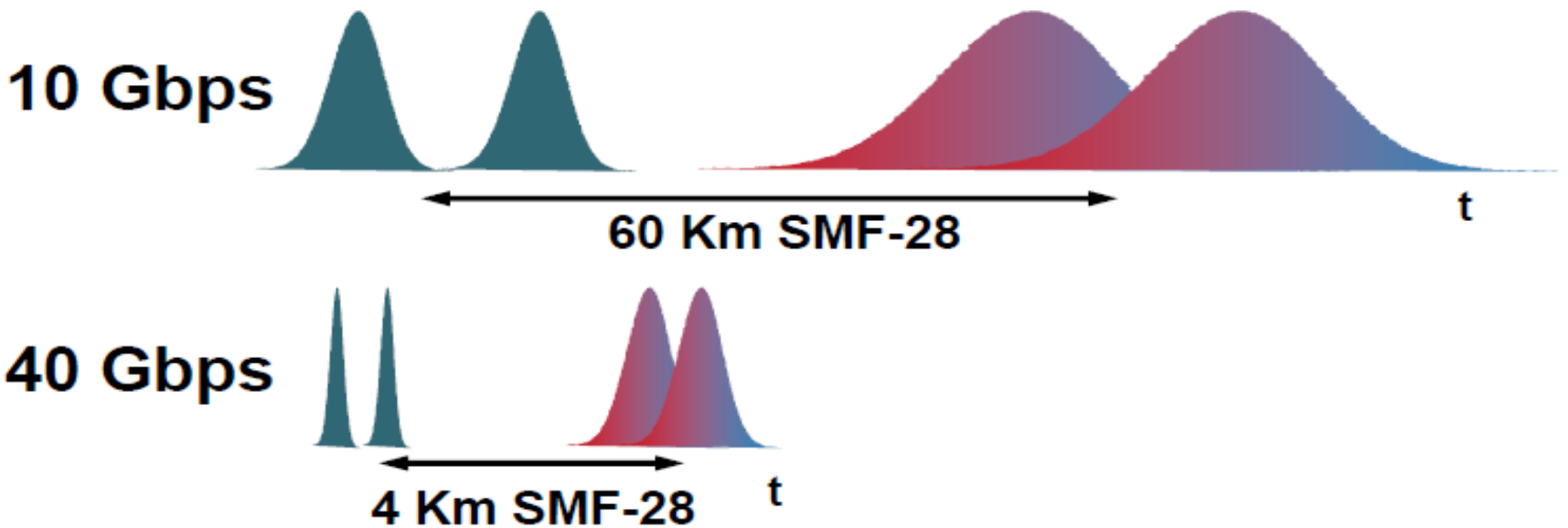
Mgmt: Management

SMF (G.652)	■ Good for TDM at 1310 nm ■ OK for TDM at 1550 ■ OK for DWDM (with Dispersion Mgmt)
DSF (G.653)	■ OK for TDM at 1310 nm ■ Good for TDM at 1550 nm ■ Bad for DWDM (C-Band)
NZDSF (G.655)	■ OK for TDM at 1310 nm ■ Good for TDM at 1550 nm ■ Good for DWDM (C + L Bands)
Extended Band (G.652.C) (Suppressed Attenuation in the Traditional Water Peak Region)	■ Good for TDM at 1310 nm ■ OK for TDM at 1550 nm ■ OK for DWDM (with Dispersion Mgmt ■ Good for CWDM (> Eight wavelengths)

The Primary Difference Is in the Chromatic Dispersion Characteristics

Limitations from Chromatic Dispersion

- Dispersion causes pulse distortion, pulse "smearing" effects
- Higher bit-rates and shorter pulses are less robust to Chromatic Dispersion
- Limits "how fast" and "how far"



Combating Chromatic Dispersion

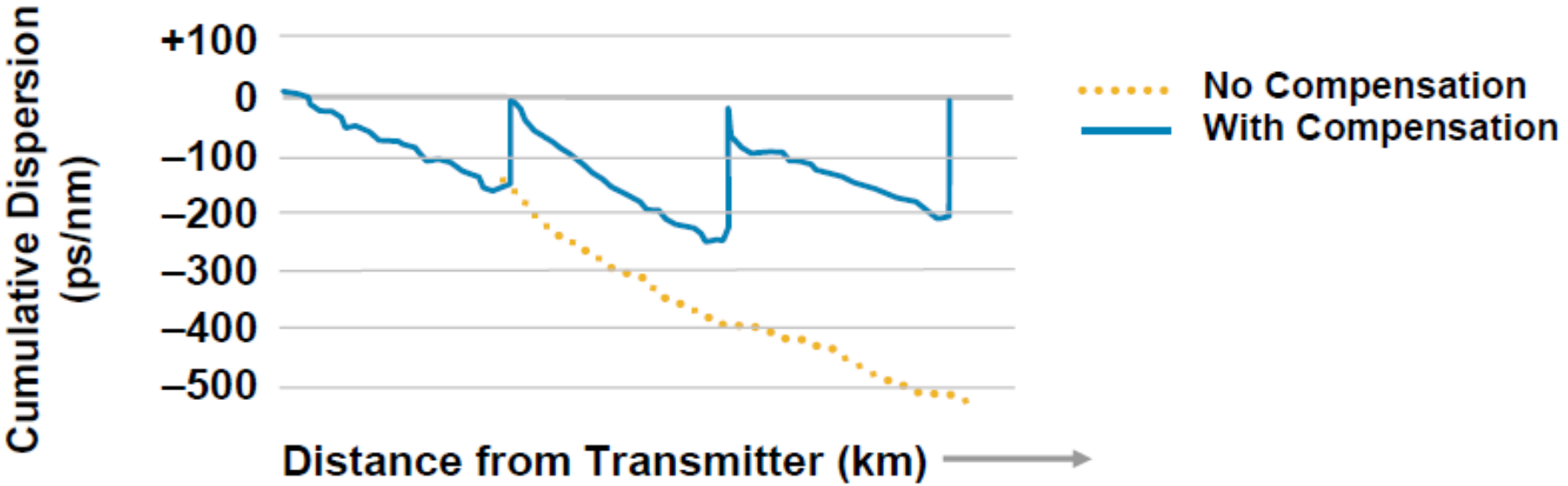
- Specialized fibers: DSF and NZDSF fibers

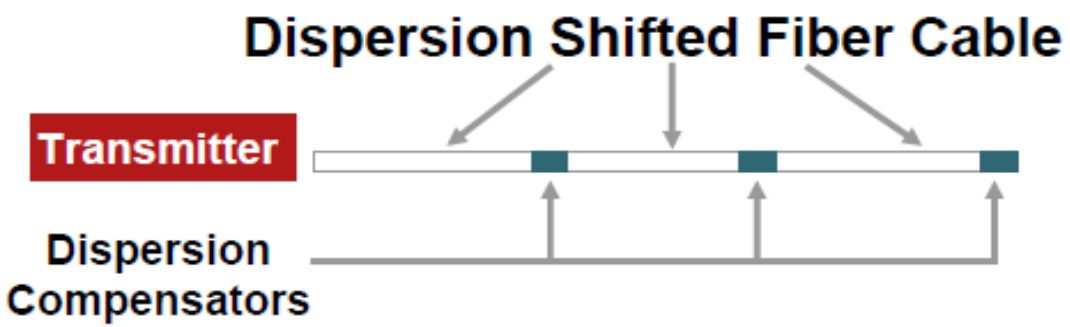
  (G.653 and G.655)

  Dispersion compensating fiber
- Transmitters with narrow spectral width
- Regenerate pulse (O-E-O)

Dispersion Compensation

Total Dispersion Controlled

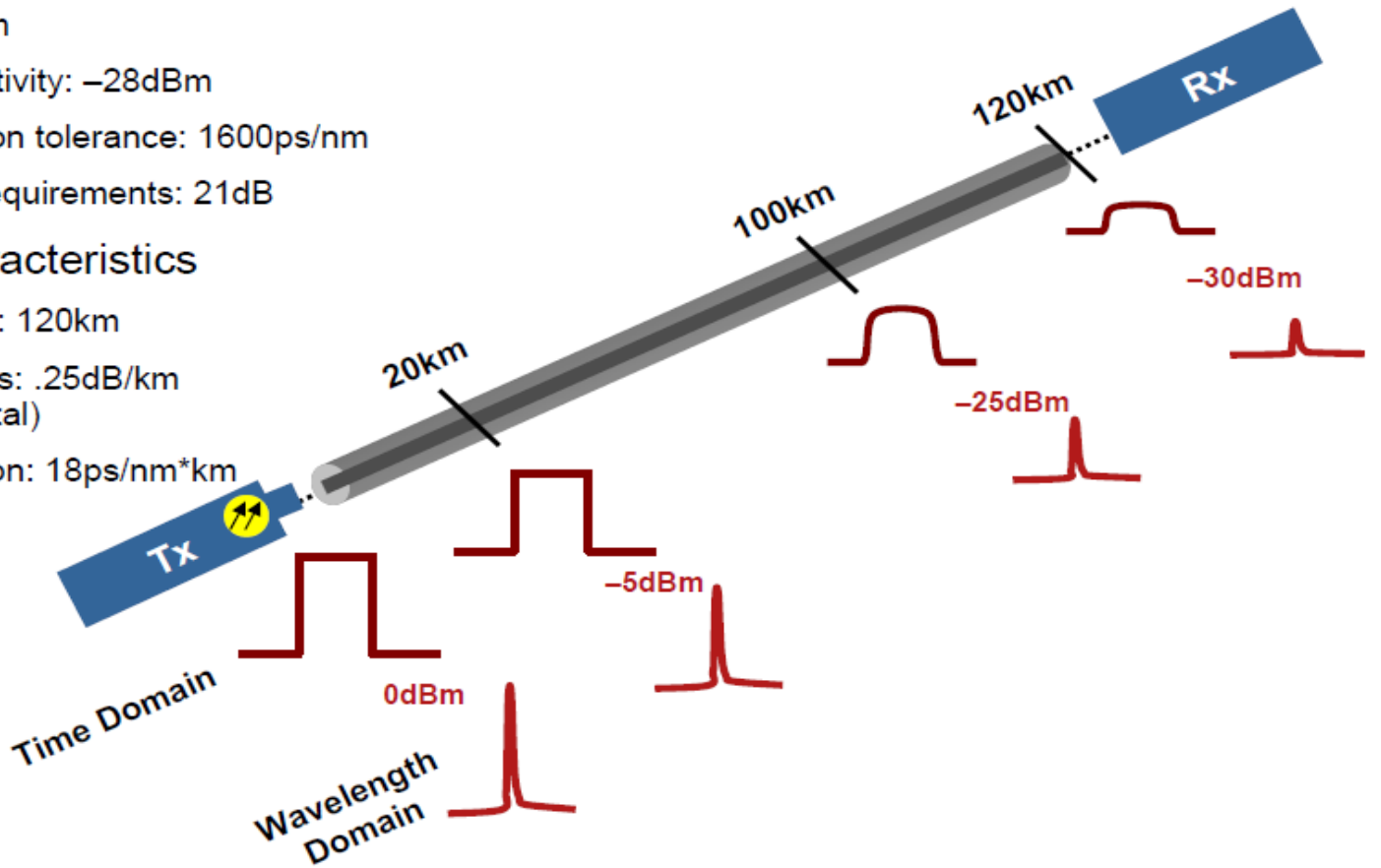




Span Design

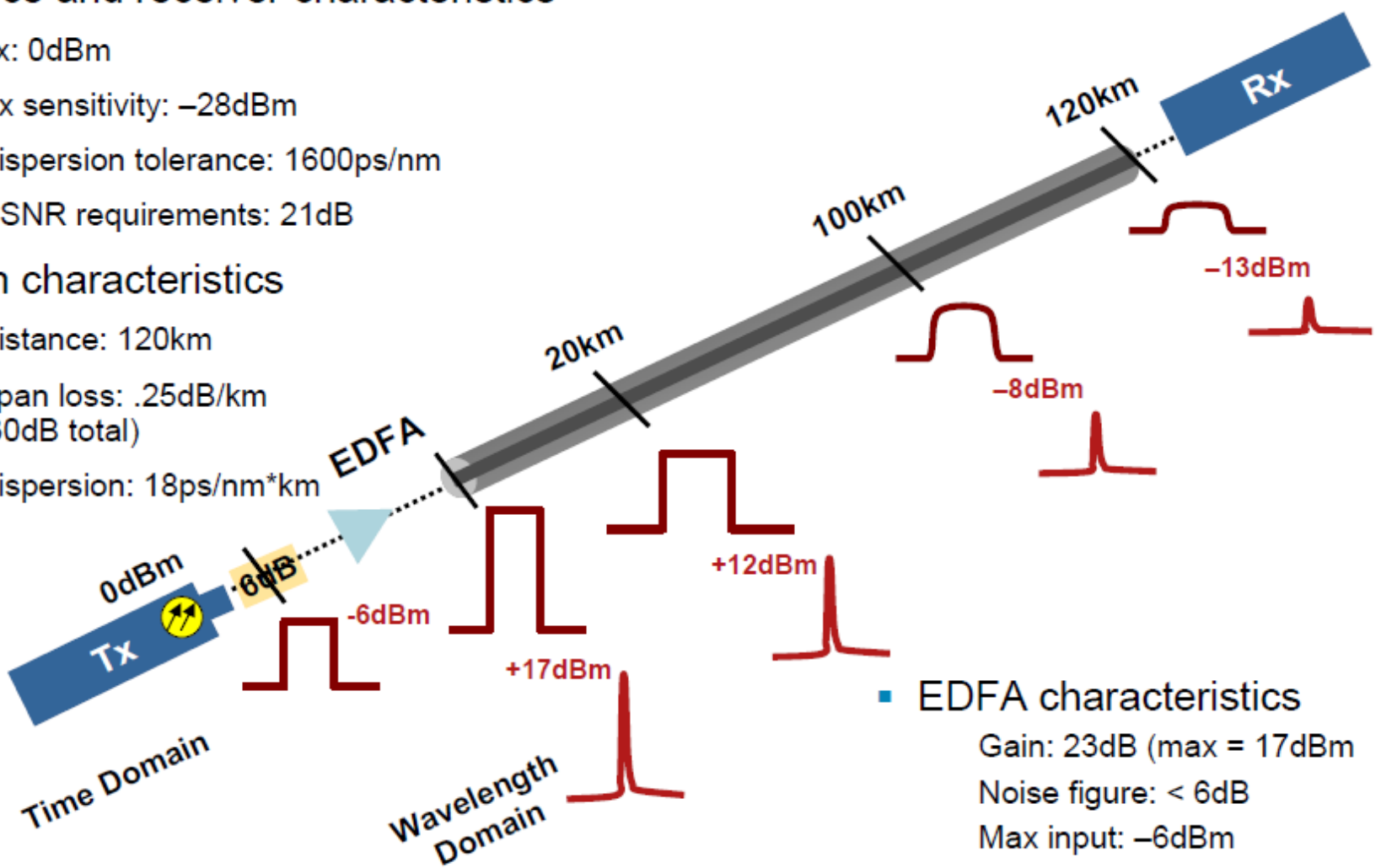
Span Design Limits Attenuation

- Source and receiver characteristics
  - Tx: 0dBm
  - Rx sensitivity: –28dBm
  - Dispersion tolerance: 1600ps/nm
  - OSNR requirements: 21dB
- Span characteristics
  - Distance: 120km
  - Span loss: .25dB/km (30dB total)
  - Dispersion: 18ps/nm*km



Span Design Limits Amplification

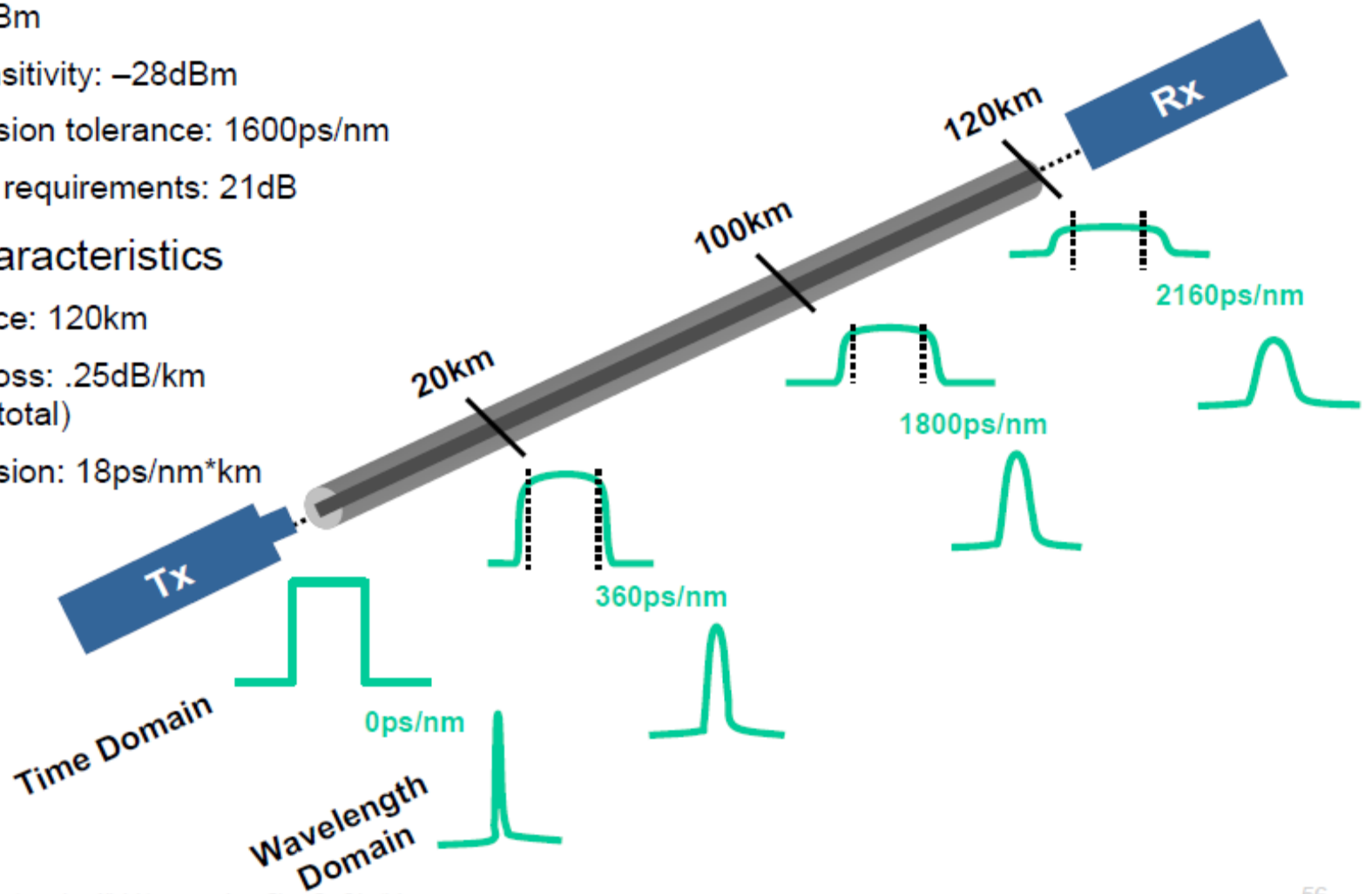
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  - Dispersion: 18ps/nm*km



- EDFA characteristics
  - Gain: 23dB (max = 17dBm
  - Noise figure: < 6dB
  - Max input: –6dBm

Span Design Limits Dispersion

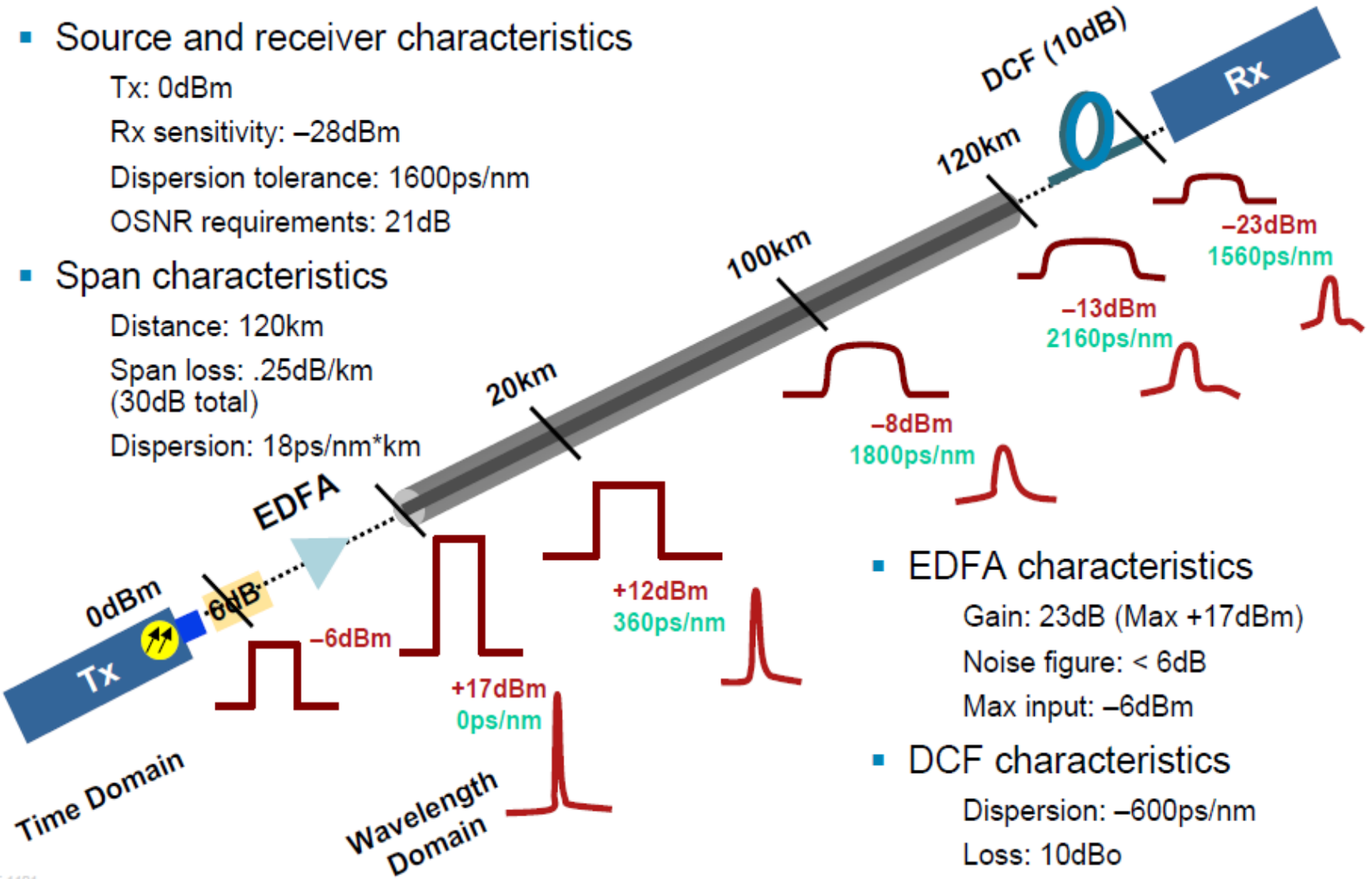
- Source and receiver characteristics
  - Tx: 0dBm
  - Rx sensitivity: –28dBm
  - Dispersion tolerance: 1600ps/nm
  - OSNR requirements: 21dB
- Span characteristics
  - Distance: 120km
  - Span loss: .25dB/km (30dB total)
  - Dispersion: 18ps/nm*km



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Span Design Limits
Dispersion Compensation

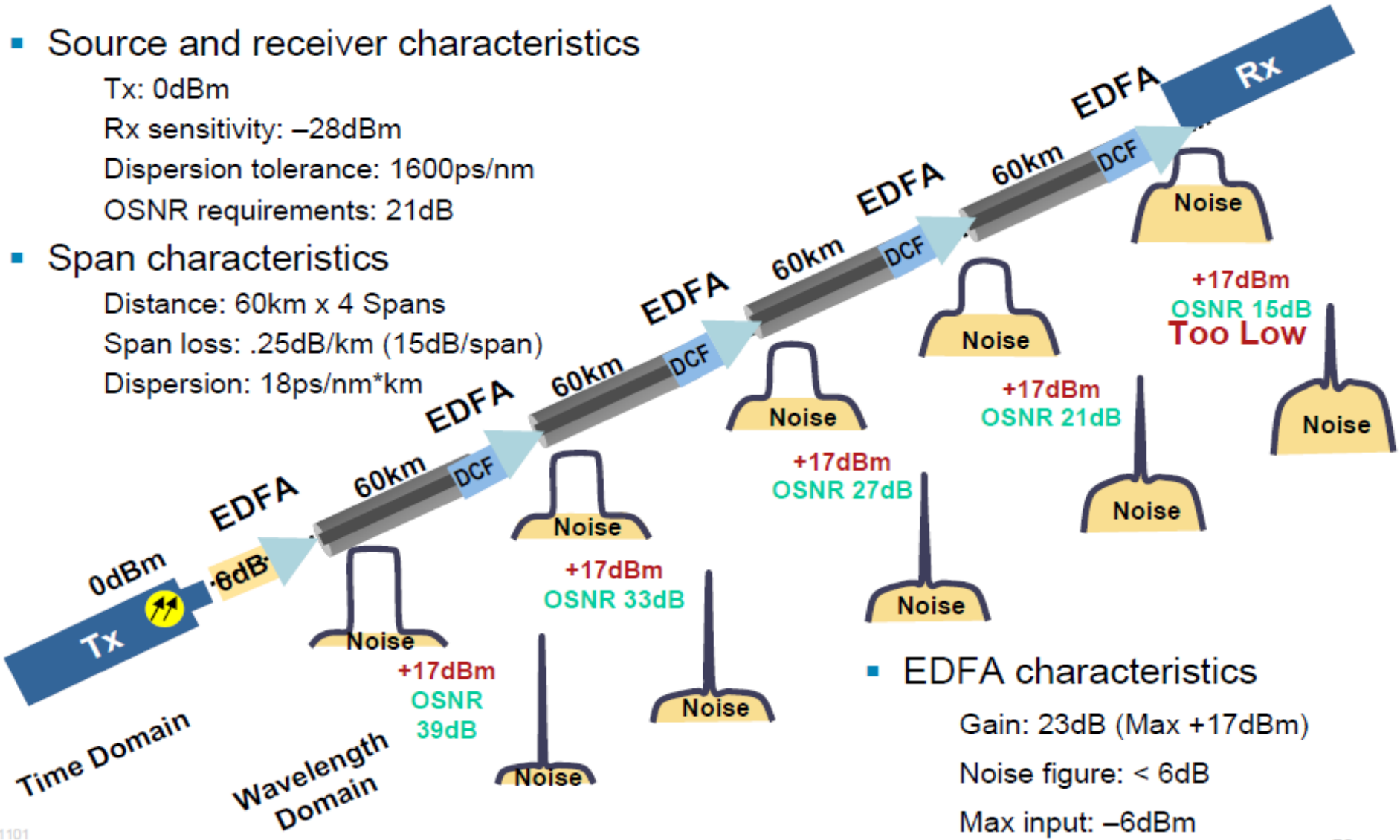
- Source and receiver characteristics
  - Tx: 0dBm
  - Rx sensitivity: –28dBm
  - Dispersion tolerance: 1600ps/nm
  - OSNR requirements: 21dB
- Span characteristics
  - Distance: 120km
  - Span loss: .25dB/km (30dB total)
  - Dispersion: 18ps/nm*km



- EDFA characteristics
  - Gain: 23dB (Max +17dBm)
  - Noise figure: < 6dB
  - Max input: –6dBm
- DCF characteristics
  - Dispersion: –600ps/nm
  - Loss: 10dBo

Span Design
Limits of Amplification (OSNR)

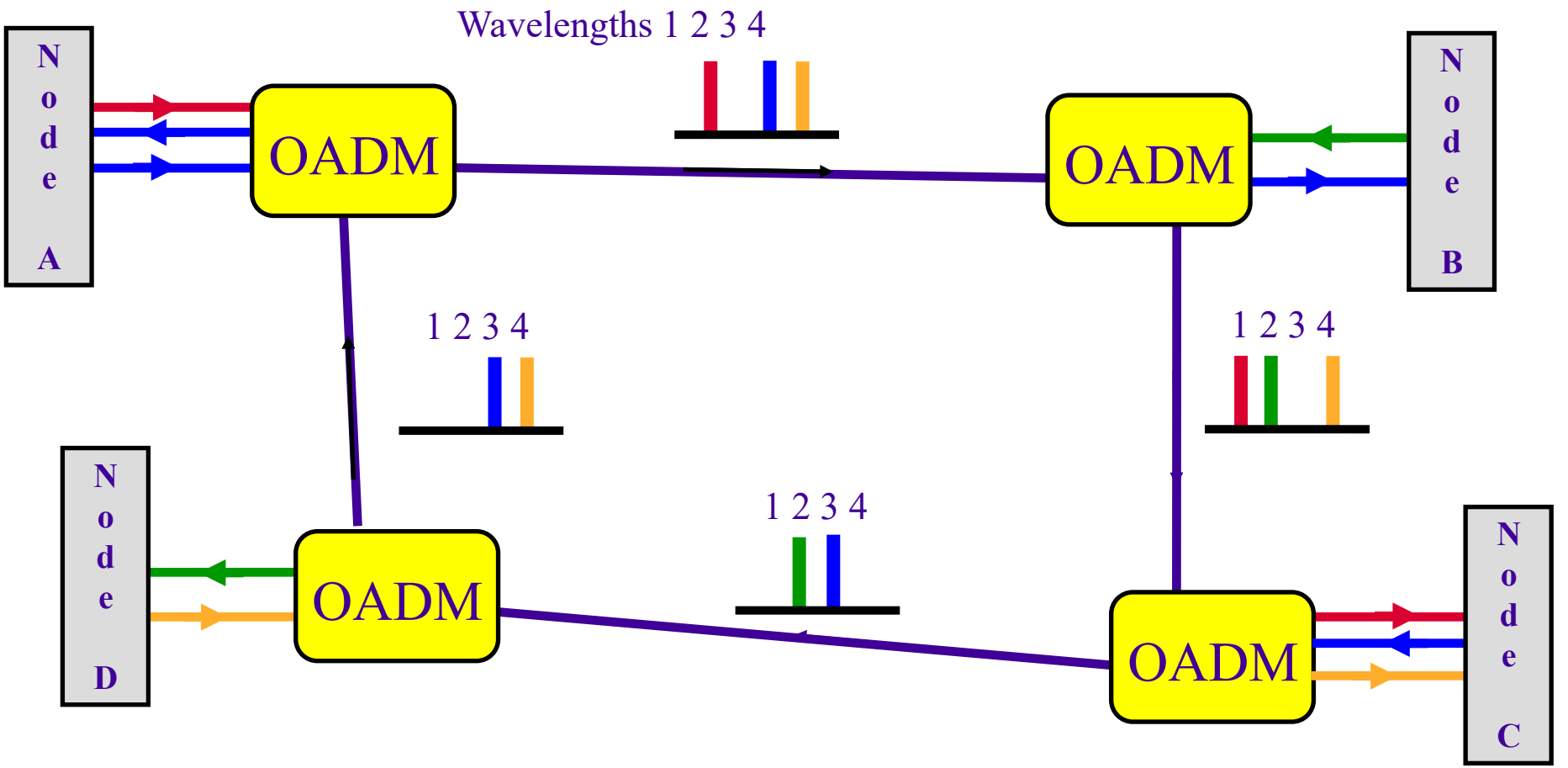
- Source and receiver characteristics
  - Tx: 0dBm
  - Rx sensitivity: –28dBm
  - Dispersion tolerance: 1600ps/nm
  - OSNR requirements: 21dB
- Span characteristics
  - Distance: 60km x 4 Spans
  - Span loss: .25dB/km (15dB/span)
  - Dispersion: 18ps/nm*km



- EDFA characteristics
  - Gain: 23dB (Max +17dBm)
  - Noise figure: < 6dB
  - Max input: –6dBm

Simple Optical Network Example

OADM: Optical Add-Drop Multiplexer

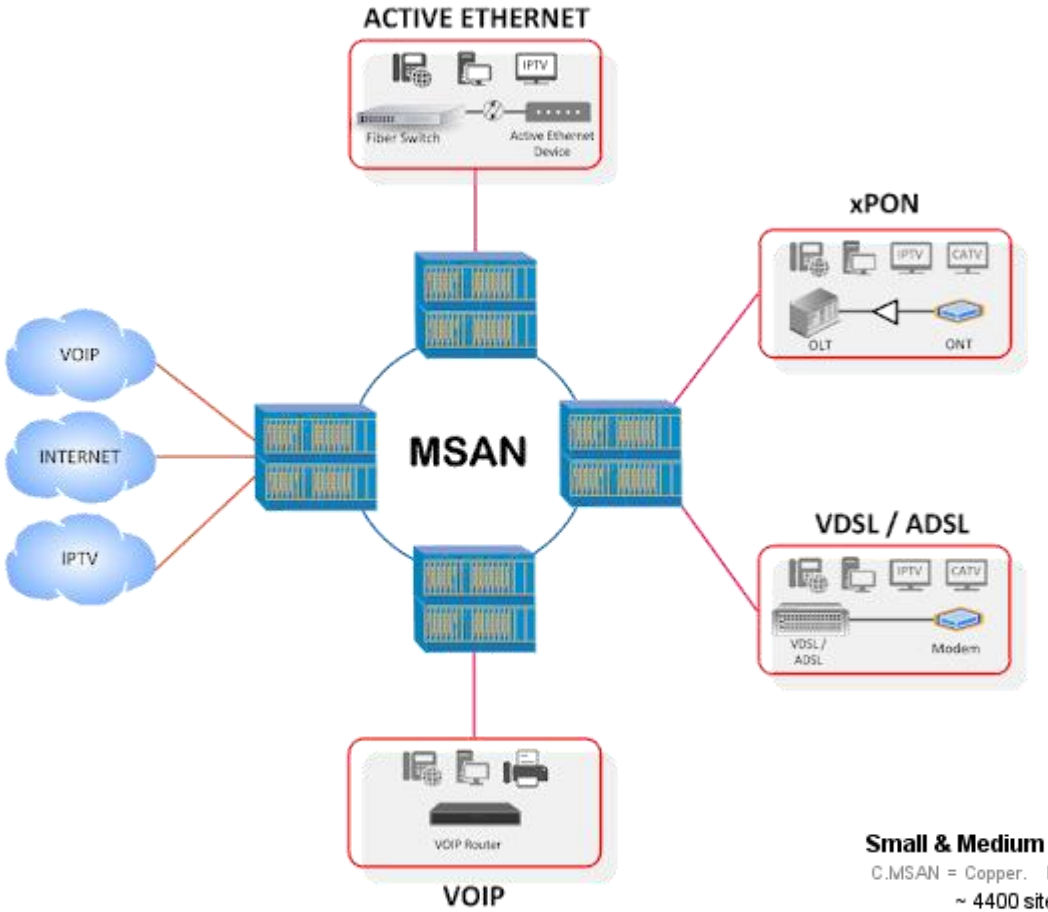


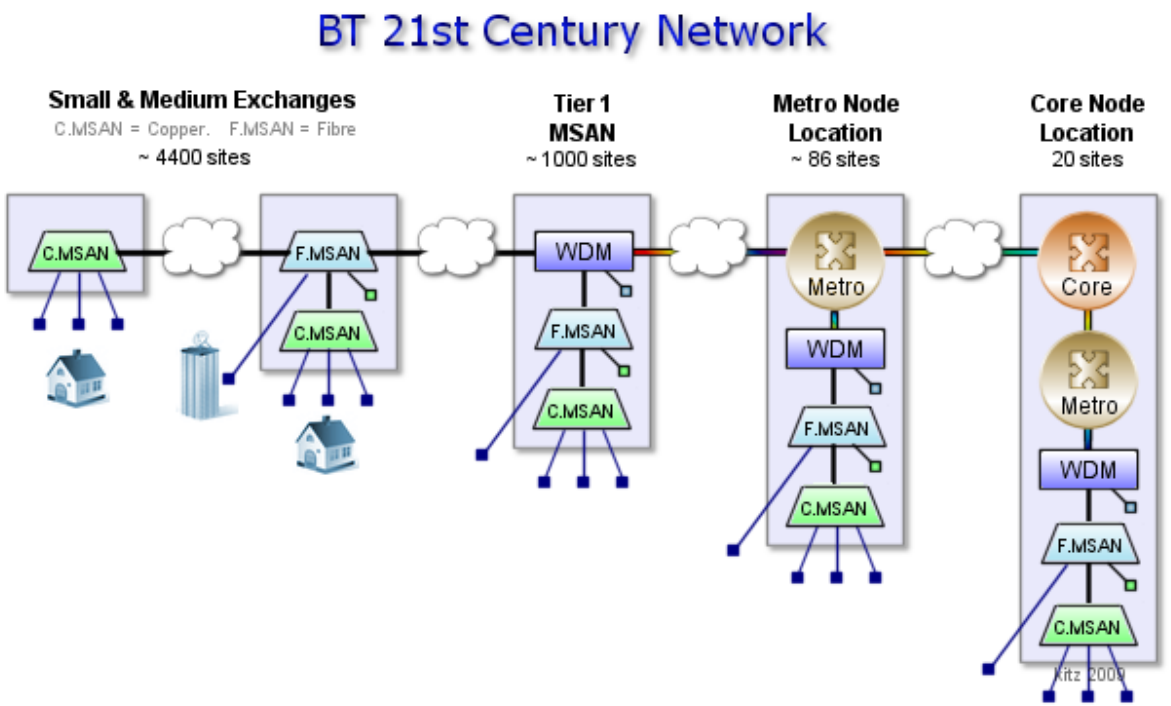
Note wavelength reuse of "blue" wavelength (no. 3), links Node A and B as well as Node C and A

Multi-Service Access Node (MSAN)

A **multi-service access node (MSAN)**, also known as a **multi-service access gateway (MSAG)**, is a device typically installed in a telephone exchange which connects customers' telephone lines to the core network, to provide telephone, ISDN, and broadband such as DSL all from a single platform.

Integrating all services on a single node, which typically backhauls all data streams over IP or Asynchronous Transfer Mode can be more cost effective and may provide new services to customers quicker than previously possible.





BT : British Telecommunications

Broadband Access Service

Digital subscriber line (DSL) is the most widely deployed broadband technology worldwide. This delivery option uses existing telephone lines to send broadband information on a different frequency than is used for the existing voice service. Many generations of DSL are used for residential service, including Very High Speed Digital Subscriber Line 2 (VDSL2) and versions of Asymmetric Digital Subscriber Line (ADSL, ADSL2, and ADSL2+). These variations of DSL primarily offer asymmetric residential broadband service where different upstream and downstream speeds are implemented. (VDSL2 also supports symmetric operation.) Other DSL variations, like High bit rate Digital Subscriber Line (HDSL) and Symmetric Digital Subscriber Line (SDSL), provide symmetric speeds and are typically used in business applications.

Broadband Access Service

Active Ethernet

Active Ethernet uses traditional Ethernet technology to deliver broadband service across a fiber-optic network. Active Ethernet does not provide a separate channel for existing voice service, so VoIP (or TDM-to-VoIP) equipment is required. In addition, sending full-speed (10 or 100 Mbps) Ethernet requires significant power, necessitating distribution to Ethernet switches and optical repeaters located in cabinets outside of the central office. Due to these restrictions, early Active Ethernet deployments typically appear in densely populated areas.

Broadband Access Service

Passive Optical Networking

Passive Optical Networking (PON), like Active Ethernet, uses fiber-optic cable to deliver services to the premises. This delivery option provides higher speeds than DSL but lower speeds than Active Ethernet. Though PON provides higher speed to each subscriber, it requires a higher investment in cable and connectivity. A key advantage of PON is that it does not require any powered equipment outside of the central office. Each fiber leaving the central office is split using a non-powered optical splitter. The split fiber then follows a point-to-point connection to each subscriber.

Broadband Access Service

PON technologies fall into three general categories:

ATM PON (APON), APON—The first passive optical network standard is primarily used for business applications.

Broadband PON (BPON), BPON—Based on APON, BPON adds wave division multiplexing (WDM), dynamic and higher upstream bandwidth allocation, and a standard management interface to enable mixed-vendor networks.

Gigabit-capable PON (GPON), GPON—The most recent PON adaptation, GPON is based on BPON but supports higher rates, enhanced security, and a choice of which Layer 2 protocol to use (ATM, Generic Equipment Model [GEM], or Ethernet).

Broadband Access Service

Ethernet PON (EPON)—Provides capabilities similar to GPON, BPON, and APON, but uses Ethernet standards. These standards are defined by the IEEE. Gigabit Ethernet PON (GEPON) is the highest speed version.

Wave Division Multiplexing PON (WDM-PON) — A nonstandard PON which, as the name implies, provides a separate wavelength to each subscriber.

Broadband Access Service

FTTx

The term *fiber to the x* (FTTx) describes how far into the network fiber-optic cabling runs before a switch to copper cabling takes place.

The following terms are used to describe the termination point of fiber-optic cable in a network:

Fiber to the Premises (FTTP), Fiber to the Home (FTTH), Fiber to the Business (FTTB)—Fiber extends all the way to the subscriber. PON is most common for residential access, although Active Ethernet can be efficiently used in dense areas such as apartment complexes. Active Ethernet is more common for delivering services to businesses.

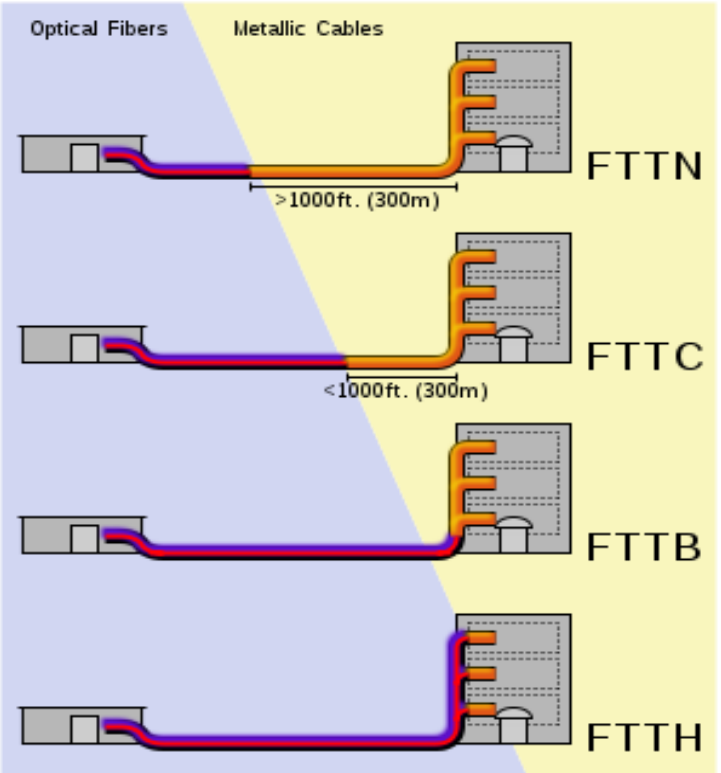
Broadband Access Service

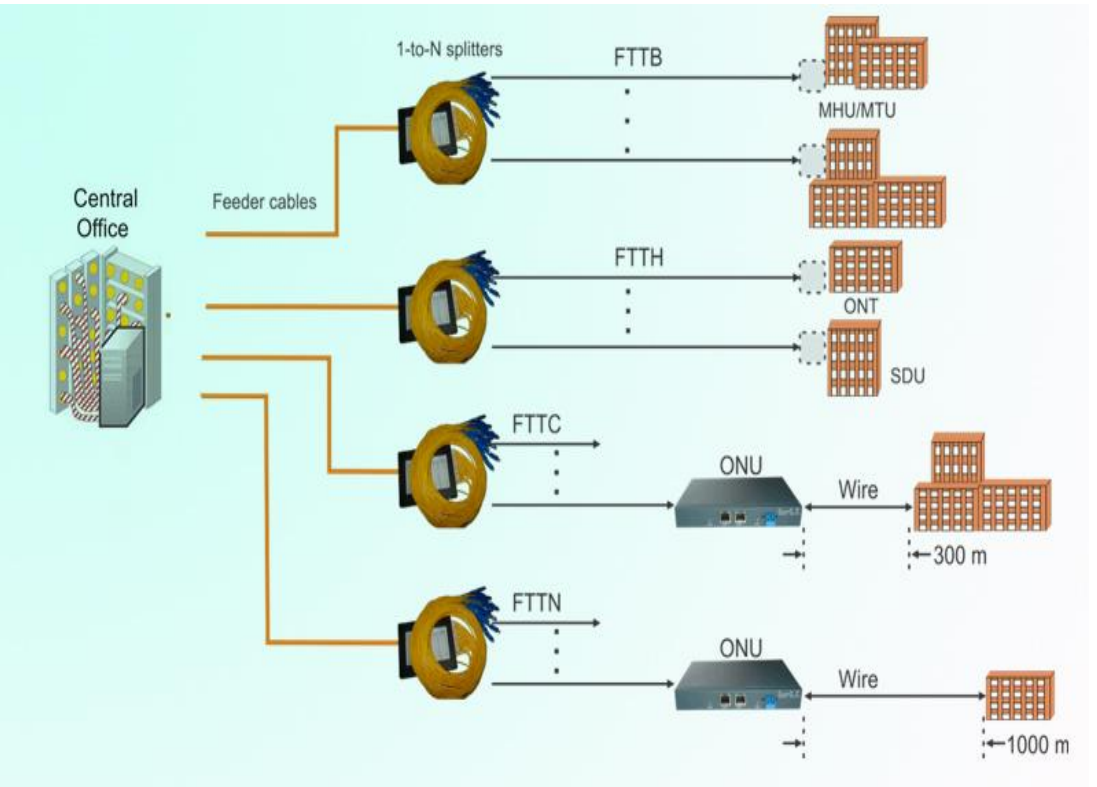
Fiber to the Curb (FTTC)—Fiber extends most of the way (typically, 500 feet/150 meters or less) to the subscriber. Existing copper is used for the remaining distance to the subscriber.

Fiber to the Node/Neighborhood (FTTN)—Fiber extends to within a few thousand feet of the subscriber and converted to xDSL for the remaining distance to the subscriber.

Fiber to the Exchange (FTTE)—A typical central office-based xDSL implementation in which fiber is used to deliver traffic to the central office and xDSL is used on the existing local loop.

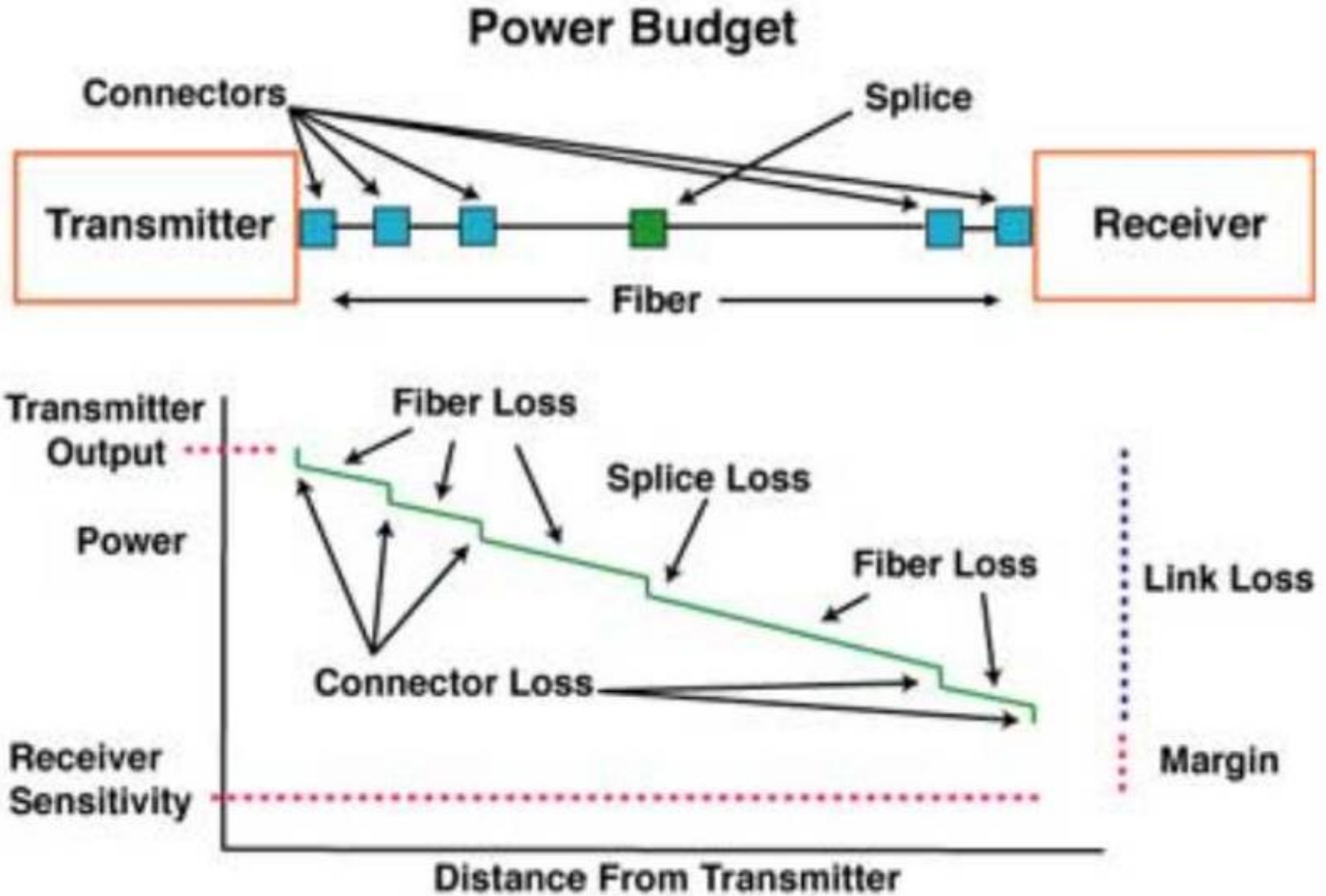
Broadband Access Service



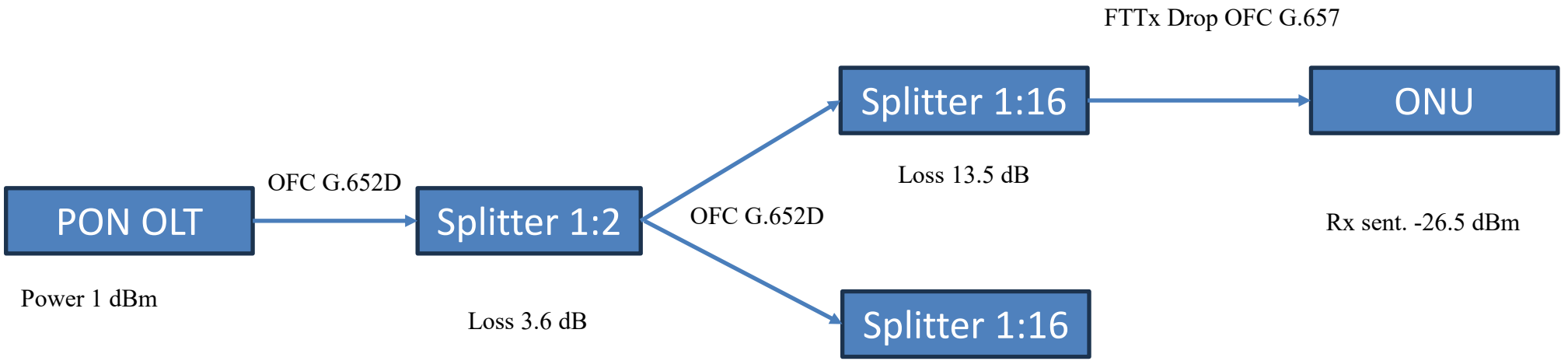


ONU (optical network units)
ONT (optical network terminal)
SDU (service data unit)
OLT (optical line terminal)

MHU/MTU (material handling unit)/(multi-tenant unit)


Power Budget
Connectors
Splice
Transmitter
Receiver
Fiber
Transmitter Output
Power
Fiber Loss
Splice Loss
Fiber Loss
Connector Loss
Link Loss
Margin
Receiver Sensitivity
Distance From Transmitter

Link Calculation แบบ Distributed Split



OFC : Optical Fiber Cable

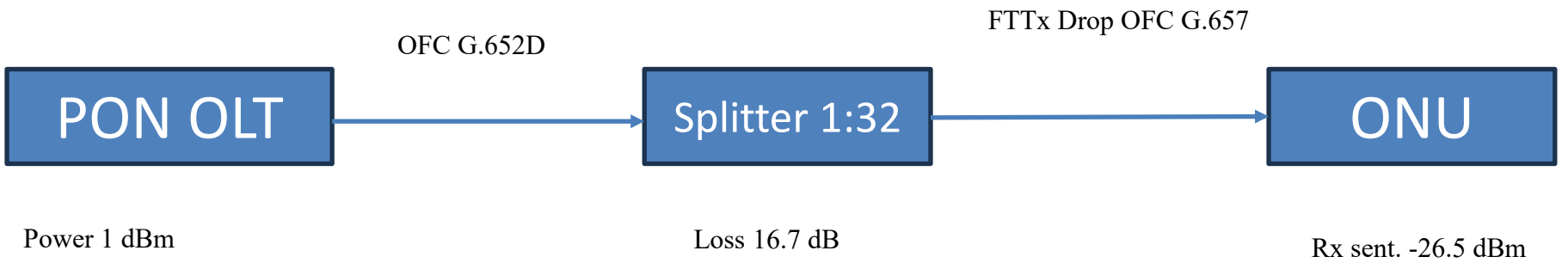
Fiber loss 0.33 dB/km @1310 nm.
OFC 5 km.
Splice loss = 1 dB (1)
Connector loss = 1 dB (2)

Fiber loss 0.33 dB/km @1310 nm.
OFC 5 km.
Splice loss = 1 dB (1)
Connector loss = 1 dB (2)

Fiber loss 0.34 dB/km @1310 nm.
OFC 500 m.
Connector loss = 1 dB (2)

Link margin >= 5 dB

Link Calculation แบบ Centralize Split



Fiber loss 0.33 dB/km @1310 nm.
OFC 5 km.
Splice loss = 1 dB (1)
Connector loss = 1 dB (2)

Fiber loss 0.34 dB/km @1310 nm.
OFC 500 m.
Connector loss = 1 dB (2)

Link margin >= 5 dB

Specification of G652D Fibre Optic Cable

Applications

- Ethernet
- Internet Protocol (IP)
- Asynchronous Transfer Mode (ATM)
- Synchronous Optical Network (SONET)
- Wavelength Division Multiplexing (WDM)

https://briticom.net/product/g652d-fibre-optic-cable/

OPTICAL CHARACTERISTICS	CONDITIONS	SPECIFIED VALUES	UNIT
Attenuation	1310nm 1383nm (after H2-aging) 1550nm 1625nm	≤0.34 ≤0.34 ≤0.20 ≤0.24	dB/km dB/km dB/km dB/km
Attenuation vs Wavelength Max. a difference	1285-1330nm, in reference to 1310nm 1525-1575nm, in reference to 1550nm	≤0.03 ≤0.02	dB/km dB/km
Dispersion Coefficient	1285-1340nm 1550nm 1625nm	-3.5 to 3.5 ≤18 ≤22	ps/(nm·km) ps/(nm·km) ps/(nm·km)
Zero Dispersion Wavelength (λσ)	–	1300-1324	nm
Zero Dispersion Slope (Sσ)	–	≤0.092	ps/(nm2·km)
Typical Value	–	0.086	ps/(nm2·km)
PMD Maximum Individual Fibre Link Design Value (M=20, Q=0.01%) Typical Value	–	≤0.1 ≤0.06 0.04	ps/√km ps/√km ps/√km
Cable Cut-Off Wavelength (λcc)	–	≤1260	nm

https://briticom.net/product/g652d-fibre-optic-cable/

Spec. G657A2 Fibre Optic Cable

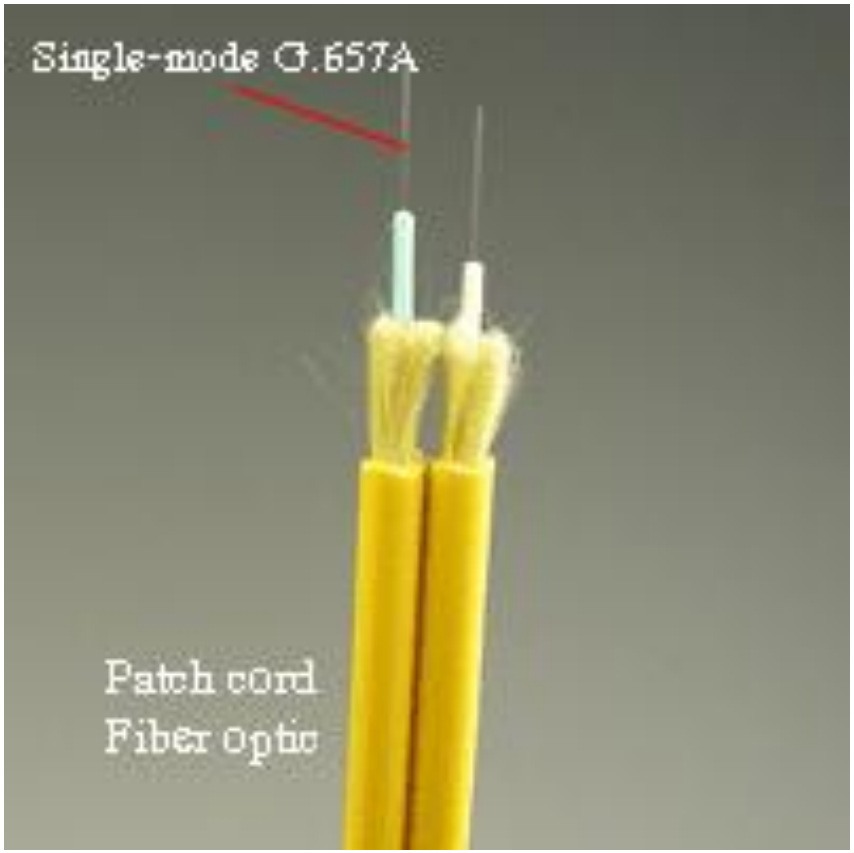
Applications

- All types of fibre cables with different structures
- High-performance optical network operating in O-E-S-C-L band
- High-speed optical routes for FTTH networks
- Cables with extreme low bending requirements
- Small-sized fibre cable and optical component

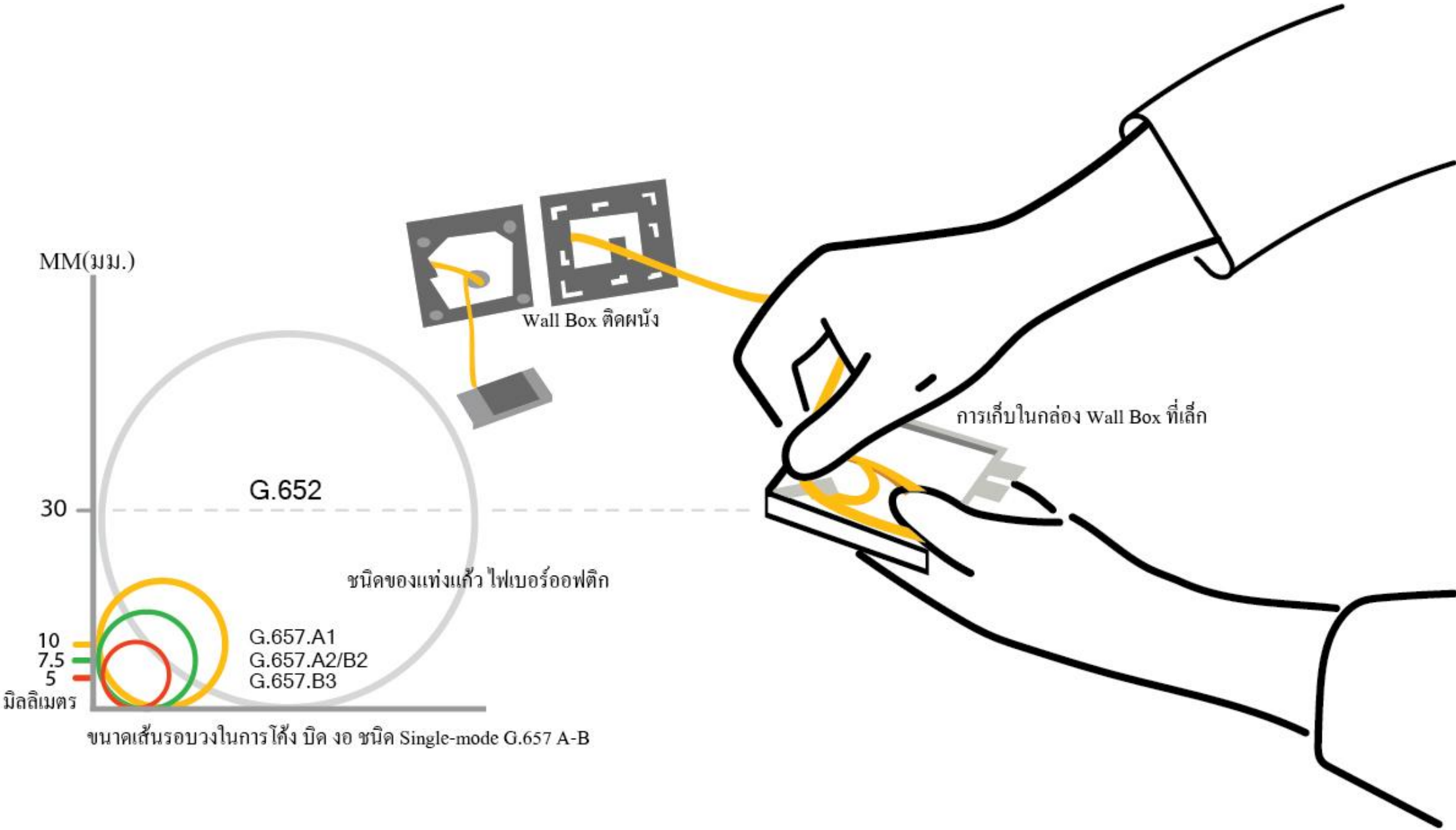
https://briticom.net/product/g657a2-fibre-optic-cable/

FEATURES | APPLICATIONS | **TECHNICAL** | DATASHEET

OPTICAL CHARACTERISTICS	CONDITIONS	SPECIFIED VALUES	UNIT
Attenuation	1310nm	≤0.35	dB/km
	1383nm (after H2-aging)	≤0.35	dB/km
	1460nm	≤0.25	dB/km
	1490nm	≤0.23	dB/km
	1550nm	≤0.21	dB/km
	1625nm	≤0.23	dB/km
Attenuation vs Wavelength	1285-1330nm, in reference to 1310nm	≤0.03	dB/km
Max. a difference	1525-1575nm, in reference to 1550nm	≤0.02	dB/km
Dispersion Coefficient	1285-1340nm	-3.5 to 3.5	ps/(nm·km)
	1550nm	≤18	ps/(nm·km)
	1625nm	≤22	ps/(nm·km)
Zero Dispersion Wavelength (λσ)	–	1300-1324	nm
Zero Dispersion Slope (Sσ)	–	≤0.092	ps/(nm2·km)
PMD		≤0.1	ps/√km
Maximum Individual Fibre	–	≤0.06	ps/√km


Single-mode G.657A
Patch cord
Fiber optic


Steel messenger wire
เส้นลวด สำหรับแขวน
LSZH sheath Jacket
เปลือกนอกผลิตจาก LSZH ชนิดพิเศษ
Optical fiber G.657.A2
แกน แท่งแก้ว
FRP Strength Member
เส้น FRP 2 เส้น รองรับแรงดึงขนานกับสาย
มอก.2166-2548


MM(มม.)
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7.5
5
มิลลิเมตร
G.652
ชนิดของแท่งแก้ว ไฟเบอร์ออฟติก
G.657.A1
G.657.A2/B2
G.657.B3
ขนาดเส้นรอบวงในการโค้ง บิด งอ ชนิด Single-mode G.657 A-B
Wall Box ติดผนัง
การเก็บในกล่อง Wall Box ที่เล็ก